

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 4: Association Analysis

Ch 4 – Association Analysis – Numerical Problems & Solutions

5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- **M1** Apriori: candidate generation, pruning, frequent itemsets
- **M2** Rule generation from frequent itemsets
- **M3** FP-tree construction and FP-Growth mining
- **M4** Pattern evaluation: lift and the correlation measures
- **M5** Subsequence enumeration

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $min_sup_count = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq min_sup_count \rightarrow L_1$.
 2. **Join** L_{k-1} with itself: combine 2 itemsets only if their first $k-2$ items match.
 3. **Prune**: discard a candidate if *any* of its $(k-1)$ -subsets is missing from L_{k-1} .
 4. **Count** survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- **Show the prune step explicitly** — it is where the method marks are.

Problem 1.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [2+7]

What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%.

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

T106 genuinely prints “F, A, C, I, B, A, P” — **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: item counts

(7 transactions)

- A:6 · P:6 · B:5 · C:5 · F:5 · M:4 · I:3 · V:3 · D:2 · L:2 · N:2 · O:2 · S:2 · others 1
- $L_1 = \{A(6), B(5), C(5), F(5), M(4), P(6)\}$

Step 2: L_2

(15 candidate pairs; counts shown, ≥ 4 kept)

- Kept: AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)
- Dropped: BC(3), BF(3), BM(2), CM(3), FM(3)

Step 3: L_3

- ABP: all 3 subsets present $\rightarrow$ counted = $3 \times$
- **ACF(4), ACP(4), AFP(4), AMP(4), CFP(4)** ✓
- Pruned without counting: ABC ($BC \notin L_2$), ABF (BF), ABM (BM), ACM (CM), AFM (FM)

Step 4: L_4

- ACFP: subsets ACF ✓ ACP ✓ AFP ✓ CFP ✓ $\rightarrow$ candidate. Count = 4 (T100, T101, T104, T106) ✓
- $L_4 = \{\mathbf{ACFP(4)}\}$. L_5 empty $\rightarrow$ stop.

Final answer

Maximal frequent itemset = **{A, C, F, P} with support 4/7 = 57%**.

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Problem 1.2 5

QUESTION AS PRINTED 2080 Baishakh · Q5 · [4+5]

Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2.

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: L_1 , count every item across the 4 transactions

- A appears in T1, T3 $\rightarrow 2$ B appears in T2, T3, T4 $\rightarrow 3$
- C appears in T1, T2, T3 $\rightarrow 3$ D appears in T1 only $\rightarrow 1$ (**dropped**)
- E appears in T2, T3, T4 $\rightarrow 3$
- $L_1 = \{\mathbf{A(2)}, \mathbf{B(3)}, \mathbf{C(3)}, \mathbf{E(3)}\}$

Step 2: C_2 , join L_1 with itself, then count each pair against the transactions

- AB: only T3 $\rightarrow 1 \times$ AC: T1, T3 $\rightarrow 2$ ✓ AE: only T3 $\rightarrow 1 \times$
- BC: T2, T3 $\rightarrow 2$ ✓ BE: T2, T3, T4 $\rightarrow 3$ ✓ CE: T2, T3 $\rightarrow 2$ ✓
- $L_2 = \{\mathbf{AC(2)}, \mathbf{BC(2)}, \mathbf{BE(3)}, \mathbf{CE(2)}\}$; dropped AB(1), AE(1)

Step 3: C_3 , join L_2 with itself, prune, then count

- BC and BE share prefix B $\rightarrow$ candidate **BCE**. Subsets BC ✓, BE ✓, CE ✓ all in $L_2 \rightarrow$ not pruned.
- AC joins with nothing else in L_2 (no other pair starts with A) $\rightarrow$ no candidate.
- Count BCE against the transactions: T2 = {B, C, E} ✓, T3 = {A, B, C, E} ✓ $\rightarrow$ count = **2** ✓ (meets threshold).
- $L_3 = \{\mathbf{BCE(2)}\}$. C_4 is empty (only one 3-itemset survived) $\rightarrow$ stop.

Final answer

Frequent itemsets: A, B, C, E, AC, BC, BE, CE, BCE.

Problem 1.3 10

QUESTION AS PRINTED 2070 Chaitra · Q4 · [10]

Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database.

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

4 transactions, 50% $\rightarrow$ min count = 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1) $\times$ AC(2) $\checkmark$ AD(1) $\times$ BC(1) $\times$ BD(2) $\checkmark$ CD(1) $\times$
- $L_2 = \{\mathbf{AC(2)}, \mathbf{BD(2)}\}$
- C_3 : AC and BD share no common prefix $\rightarrow$ **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Problems 1.4 and 1.5 5+3 8

QUESTION AS PRINTED 2076 Chaitra · Q7 · [5+3]

Consider the transaction data shown in the following table from a fast food restaurant.

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1–M5). Assume that the minimum support is 2/9 and the minimum confidence is 7/9.

- Apply the Apriori algorithm to the dataset of transactions and identify all frequent k -itemsets.
- Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

QUESTION AS PRINTED 2074 Ashwin · Q5 · [8]

Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm.

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

Take minimum support = 20%, minimum confidence 80%

Identical 9-transaction table, and both thresholds work out to a count of 2, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

- **Step 1**, L_1 : M1 in T1,T4,T5,T7,T8,T9 $\rightarrow$ 6. M2 in T1,T2,T3,T4,T6,T8,T9 $\rightarrow$ 7. M3 in T3,T5,T6,T7,T8,T9 $\rightarrow$ 6. M4 in T2,T4 $\rightarrow$ 2. M5 in T1,T8 $\rightarrow$ 2.

- $L_1 = \{M1(6), M2(7), M3(6), M4(2), M5(2)\}$ (nothing dropped, all 5 items clear count 2)
- **Step 2**, $C_2 \rightarrow L_2$: check each pair against the table. M1M2: T1,T4,T8,T9 $\rightarrow 4 \checkmark$. M1M3: T5,T7,T8,T9 $\rightarrow 4 \checkmark$. M1M4: T4 only $\rightarrow 1 \times$. M1M5: T1,T8 $\rightarrow 2 \checkmark$. M2M3: T3,T6,T8,T9 $\rightarrow 4 \checkmark$. M2M4: T2,T4 $\rightarrow 2 \checkmark$. M2M5: T1,T8 $\rightarrow 2 \checkmark$. M3M4: never together $\rightarrow 0 \times$. M3M5: T8 only $\rightarrow 1 \times$. M4M5: never together $\rightarrow 0 \times$.
- $L_2 = \{M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)\}$; dropped M1M4(1), M3M4(0), M3M5(1), M4M5(0)
- **Step 3**, C_3 : join pairs sharing a prefix, prune by subset, then count. M1M2M3 (from M1M2 + M1M3, needs M2M3 too $\rightarrow$ present) $\rightarrow$ count in T8,T9 $\rightarrow 2 \checkmark$. M1M2M5 (from M1M2 + M1M5, needs M2M5 $\rightarrow$ present) $\rightarrow$ count in T1,T8 $\rightarrow 2 \checkmark$. M1M2M4 pruned (subset M1M4 $\notin L_2$). M2M3M5 pruned (subset M3M5 $\notin L_2$). M1M3M5 pruned (subset M3M5 $\notin L_2$).
- $L_3 = \{M1M2M3(2), M1M2M5(2)\}$
- **Step 4**, C_4 : only candidate is M1M2M3M5 (from M1M2M3 + M1M2M5), pruned because its subset M1M3M5 $\notin L_3 \rightarrow L_4$ empty $\rightarrow$ stop.

1.4 – rules $X \wedge Y \Rightarrow Z$, min conf $7/9 = 0.778$

- From M1M2M3 (sup 2): all three give $2/4 = 0.5 \times$
- From M1M2M5 (sup 2):
 - $M1 \wedge M2 \Rightarrow M5 = 2/4 = 0.5 \times$
 - $M1 \wedge M5 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
 - $M2 \wedge M5 \Rightarrow M1 = 2/2 = 1.0 \checkmark$

Answer

2 strong rules, both at 100% confidence.

1.5 – all rules, min conf 0.80

- $M5 \Rightarrow M1 = 2/2 = 1.0 \checkmark$ $M5 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
- $M4 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
- $M5 \Rightarrow M1 \wedge M2 = 2/2 = 1.0 \checkmark$
- Plus the two 3-itemset rules from 1.4.
- All rules with M1, M2 or M3 as antecedent fail (best is $4/6 = 0.667$).
- **Pattern**: only the *rare* items (M4, M5) produce confident rules – they never appear without their partners.

Problem 1.6 8

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8]

Derive association rule for the following market basket transactions.

Minimum support = 50% Minimum confidence = 80%

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

This one has a surprise.

- 6 transactions, 50% $\rightarrow$ min count = 3.
- L_1 : A(4), B(3), E(3) (C:2 $\times$, D:2 $\times$)
- C_2 : AB = 1 $\times$ AE = 1 $\times$ BE = 2 $\times$
- $L_2 = \emptyset$

Final answer

- **No frequent 2-itemset exists, so no association rule can be derived at 50% support.** Say this explicitly – it is the correct answer, not a dead end.
- **Show you understand why**: the data is spread thin, every pair occurs at most twice.
- **If min support is relaxed to 33.3%** (count 2): $L_2 = \{AC(2), BE(2)\}$, and the only rule clearing 80% confidence is $C \Rightarrow A$ at $2/2 = 100\%$. Worth adding as a one-line remark. verified/added

M2. Rule generation from frequent itemsets

Method

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.

- **The support of every rule from l is the same** ($= \text{sup}(l)$); only confidence varies.
- **Shortcut:** if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a *subset* of s also fails. Prune downward.

Worked example

PROBLEM not a PYQ · the rule half of 2081 Baishakh Q3 (b) verified/added

The frequent itemset $l = \{K, E, O\}$ has support count 3 in the 5-transaction database of Problem 3.1, where $\text{sup}(K) = 5$, $\text{sup}(E) = 4$, $\text{sup}(O) = 3$, $\text{sup}(K, E) = 4$, $\text{sup}(K, O) = 3$ and $\text{sup}(E, O) = 3$. List every association rule that l can generate and its confidence, and say which are strong at a minimum confidence of 80%.

From $l = \{K, E, O\}$ with $\text{sup} = 3$ (see Problem 3.1), $|l| = 3 \rightarrow 6$ candidate rules:

Rule	Confidence	Value	$\geq 80\%$?
$K \Rightarrow E, O$	$3/5$	0.60	×
$E \Rightarrow K, O$	$3/4$	0.75	×
$O \Rightarrow K, E$	$3/3$	1.00	✓
$K, E \Rightarrow O$	$3/4$	0.75	×
$K, O \Rightarrow E$	$3/3$	1.00	✓
$E, O \Rightarrow K$	$3/3$	1.00	✓

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

M3. FP-tree construction and FP-Growth mining

Method

1. **Scan 1:** count items, drop infrequent ones, sort the rest by descending support $\rightarrow$ **F-list**. **State your tie-break rule.**
2. **Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
3. **Mine bottom-up** (least frequent item first): conditional pattern base $\rightarrow$ conditional FP-tree $\rightarrow$ recurse.

Problem 3.1 2+7

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [2+7]

What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%.

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

- i) Find all frequent itemsets using FP-growth.
 - ii) List all of the strong association rules (with support and confidence).
- **T500 prints O twice** — a transaction is a set, so it is {C,O,K,I,E}.
 - 5 transactions, 60% $\rightarrow$ **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 ✓ N:2, C:2, D:1, A:1, U:1, I:1 ×
- **F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

- root
- → K:5
- → E:4
- → M:2 → O:1 → Y:1
- → O:2 → Y:1
- → M:1 → Y:1
- **Check:** counts on each level sum correctly – K:5 = E:4 + M:1. ✓

All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)

- $K \Rightarrow E = 4/5 = 0.80$ ✓ $E \Rightarrow K = 4/4 = 1.0$ ✓
- $M \Rightarrow K = 3/3 = 1.0$ ✓ $O \Rightarrow K = 3/3 = 1.0$ ✓ $Y \Rightarrow K = 3/3 = 1.0$ ✓ $O \Rightarrow E = 3/3 = 1.0$ ✓
- $K, O \Rightarrow E = 1.0$ ✓ $E, O \Rightarrow K = 1.0$ ✓ $O \Rightarrow K, E = 1.0$ ✓
- Failing: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ (all $3/5 = 0.6$), $E \Rightarrow O$ (0.75), $K, E \Rightarrow O$ (0.75), $E \Rightarrow K, O$ (0.75).

Problem 3.2 4+4

QUESTION AS PRINTED 2075 Chaitra · Q4 · [4+4]

What do you mean by frequent Pattern growth, draw FP-tree with given tabular data.

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

Min support is not printed; use 3, which is the standard for this dataset. verified/added**Counts and F-list**

- f:4, c:4, a:3, b:3, m:3, p:3 ✓ l:2, o:2, and d,g,i,h,j,k,s,e,n,w all 1 ×
- **F-list = f, c, a, b, m, p**
- **T05 prints a capital "A"** – same item as a.

Reordered transactions

- 01 → f, c, a, m, p 02 → f, c, a, b, m
- 03 → f, b 04 → c, b, p
- 05 → f, c, a, m, p

Step 4: mine bottom-up

Item	Conditional pattern base	Frequent patterns
Y	{K,E,M:1}, {K,E:1}, {K,M:1}	{K,Y}:3
O	counts K:3, E:2, M:2 {K,E,M:1}, {K,E:2}	{K,O}:3, {E,O}:3,
M	counts K:3, E:3 {K,E:2}, {K:1}	{K,E,O}:3 {K,M}:3
E	counts K:3, E:2 {K:4}	{K,E}:4
K	–	{K}:5

The FP-tree

- root
- → f:4
- → c:3 → a:3
- → m:2 → p:2
- → b:1 → m:1
- → b:1
- → c:1 → b:1 → p:1
- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

QUESTION AS PRINTED 2078 Bhadra · Q5 · [4+4]

Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following:

- Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
- Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- 6 transactions, 33.34% → **min count = 2**.
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent**.
- F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2)** (4-way tie broken alphabetically).

Tree, built transaction by transaction

- T1 (HotDogs, Buns, Ketchup): root → H:1 → Bu:1 → Ke:1
- T2 (HotDogs, Buns): H:2, Bu:2
- T3 (Chips, HotDogs, Coke): new branch
Ch:1 → H:1 → Co:1
- T4 (Chips, Coke): Ch:2, new child Co:1
- T5 (Chips, Ketchup): Ch:3, new child Ke:1
- T6 (Chips, HotDogs, Coke): Ch:4, H:2, Co:2

Final tree

- root
- Chips:4 → {HotDogs:2 → Coke:2, Coke:1, Ketchup:1}
- HotDogs:2 → Buns:2 → Ketchup:1

Mining

Item	Cond. pattern base	Patterns
Ketchup	{H,Bu:1}, {Ch:1}	none (all counts 1)
Buns	{H:2}	{H,Bu:2}
Coke	{Ch,H:2}, {Ch:1}	{Ch,Co:3}, {H,Co:2}, {Ch,H,Co:2}
HotDogs	{Ch:2}	{Ch,H:2}
Chips	—	{Ch:4}

Frequent itemsets

- Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- {Chips,Coke} 3, {Chips,HotDogs} 2, {HotDogs,Coke} 2, {HotDogs,Buns} 2
- {Chips,HotDogs,Coke} 2

Problem 3.4 2+8

QUESTION AS PRINTED 2074 Chaitra · Q6 · [2+8]

How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree.

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

Min support is not printed; use 2.

- Counts: Laptop 4, Earphone 4, Mobile 3, Camera 2, Pen drive 2 — **all frequent**.
- F-list = Earphone(4), Laptop(4), Mobile(3), Camera(2), Pen drive(2)**
- Reordered: T1 (L, Ca, Pd) · T2 (L, Pd) · T3 (E, L, Mo) · T4 (E, Mo) · T5 (E, Ca) · T6 (E, L, Mo)
- Tree:** root → Earphone:4 → {Laptop:2 → Mobile:2, Mobile:1, Camera:1}; and root → Laptop:2 → {Camera:1 → PenDrive:1, PenDrive:1}
- Key frequent pattern: **{Earphone, Laptop, Mobile}:2**

M4. Pattern evaluation: lift and correlation

Method

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \Rightarrow B)}{sup(B)} \quad \chi^2 = \sum \frac{(obs - exp)^2}{exp}$$

- $lift > 1$ positive · $= 1$ independent · < 1 negative.

Problem 4.1 8

QUESTION AS PRINTED 2070 Chaitra · Q5 · [8]

Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules.

The paper prints no data — the example is yours to supply. The standard *game* / *video* contingency table below is the one every textbook uses. verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $sup(game \Rightarrow video) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $conf(game \Rightarrow video) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(video) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$lift = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<i>game</i>	row
video	4000 (4500)	3500 (3000)	7500
<i>video</i>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer **all_conf**, **max_conf**, **Kulczynski** or **cosine**.

M5. Subsequence enumeration

Method

- A ***k*-subsequence** contains *k* items in total (not *k* elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (*a, b, c, d*) of items taken per element, then remove duplicates.

Problem 5.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (a) · [3]

List all the 4-subsequences contained in the data sequence: $\langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

Step 1: enumerate the item-count splits

(a, b, c, d) with $a \leq 2, b \leq 1, c \leq 2, d \leq 1$, $\text{sum} = 4$

(a, b, c, d)	Ways	Subsequences
(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Final answer

- 15 selections, but $\langle \{1, 3\} \{2\} \{4\} \rangle$ arises twice (the $\{2\}$ can come from e_2 or e_3), so there are **14 distinct 4-subsequences**.
- **The exam usually only wants “at least four”** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Problem 5.2 3

QUESTION AS PRINTED 2079 Bhadra · Q6 (a) · [3+3]

How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$.

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- **Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.